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Task : 2

Exploring User Accounts, Permissions, and Access Control

- Linux supports multiple user accounts, each with specific access rights.
- User information is stored in /etc/passwd, and password-related data is stored securely in /etc/shadow.
- Access control ensures users can only access files and resources they are authorized to use.
- This reduces the risk of unauthorized access and privilege misuse.

Understanding File Permissions Using chmod, chown, and ls -l

Definition

- File permissions control who can read, write, or execute a file or directory in Linux.

Explanation

- Linux uses three permission types:
 - **Read (r)** – view file content
 - **Write (w)** – modify file content
 - **Execute (x)** – run the file as a program
- Permissions are applied to three users:
 - Owner
 - Group
 - Others

Example

- `chmod 755 file.sh`
 - Owner can read, write, execute
 - Group and others can read and execute only

Administrator vs Standard User Privileges

- The **root (administrator)** user has complete control over the system.
- A **standard user** has limited permissions and cannot make critical system changes.
- Using standard accounts for daily work reduces the impact of malware and accidental damage.
- Administrative access should be used only when required.

Key point

- Root is powerful but dangerous.
- Normal users protect the system by default.

Enabling Firewall (UFW or Windows Firewall)

- A firewall monitors and controls incoming and outgoing network traffic.
- In Linux, **UFW (Uncomplicated Firewall)** is enabled to block unauthorized connections.
- In Windows, **Windows Defender Firewall** provides similar protection.
- Firewalls are a basic but essential layer of OS security.

Identifying Running Processes and Services

- Running processes show what programs are currently active on the system.
- Commands like ps, top, and htop help monitor system activity in Linux.
- On Windows, Task Manager displays running processes and services.
- Monitoring processes helps detect suspicious or unwanted activity.

Disabling Unnecessary Services to Reduce Attack Surface

- Unused services increase the system's attack surface.
- Attackers often exploit unnecessary background services.
- Disabling unused services improves performance and security.
- Only essential services should be allowed to run.

Best OS Hardening Practices

- Keep the operating system updated with the latest patches.
- Use strong passwords and enable account lockout policies.
- Apply the **principle of least privilege**.
- Enable firewalls and antivirus protection.
- Disable unnecessary services and ports.

- Regularly audit users, permissions, and logs.

Final Outcome

- This task helps build a strong understanding of **OS-level security and hardening**.
- It demonstrates how proper configuration reduces security risks at the operating system level.

Deliverable: OS Security Checklist

You can directly paste this into your **OS security checklist document**.

Linux / Windows OS Hardening Checklist

- Installed OS from a trusted source
- Created standard user accounts (not using admin daily)
- Used strong passwords
- Applied latest security updates
- Verified user accounts and removed unused ones
- Configured file permissions properly
- Used chmod, chown, and ls -l (Linux)
- Enabled firewall:
 - UFW on Linux
 - Windows Defender Firewall on Windows
- Checked running processes and services
- Disabled unnecessary services
- Limited root or administrator access
- Followed least privilege principle